

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 15



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Recap of Previous Lecture



Topic

Complements of an element in a lattice



Topic

Sub-lattice



Topics to be Covered



Topic

Distributive lattice





Topic : Distributive lattice

Let $[L, \vee, \wedge]$ be a lattice.

$[L, \vee, \wedge]$ is a distributive lattice if and only if,

$$(i) \quad a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$\text{And (ii) } a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \quad \leftarrow \forall a, b, c \in L$$

Lattice is called a distributive lattice if and only if distributive property is satisfied by every combination of elements of the lattice

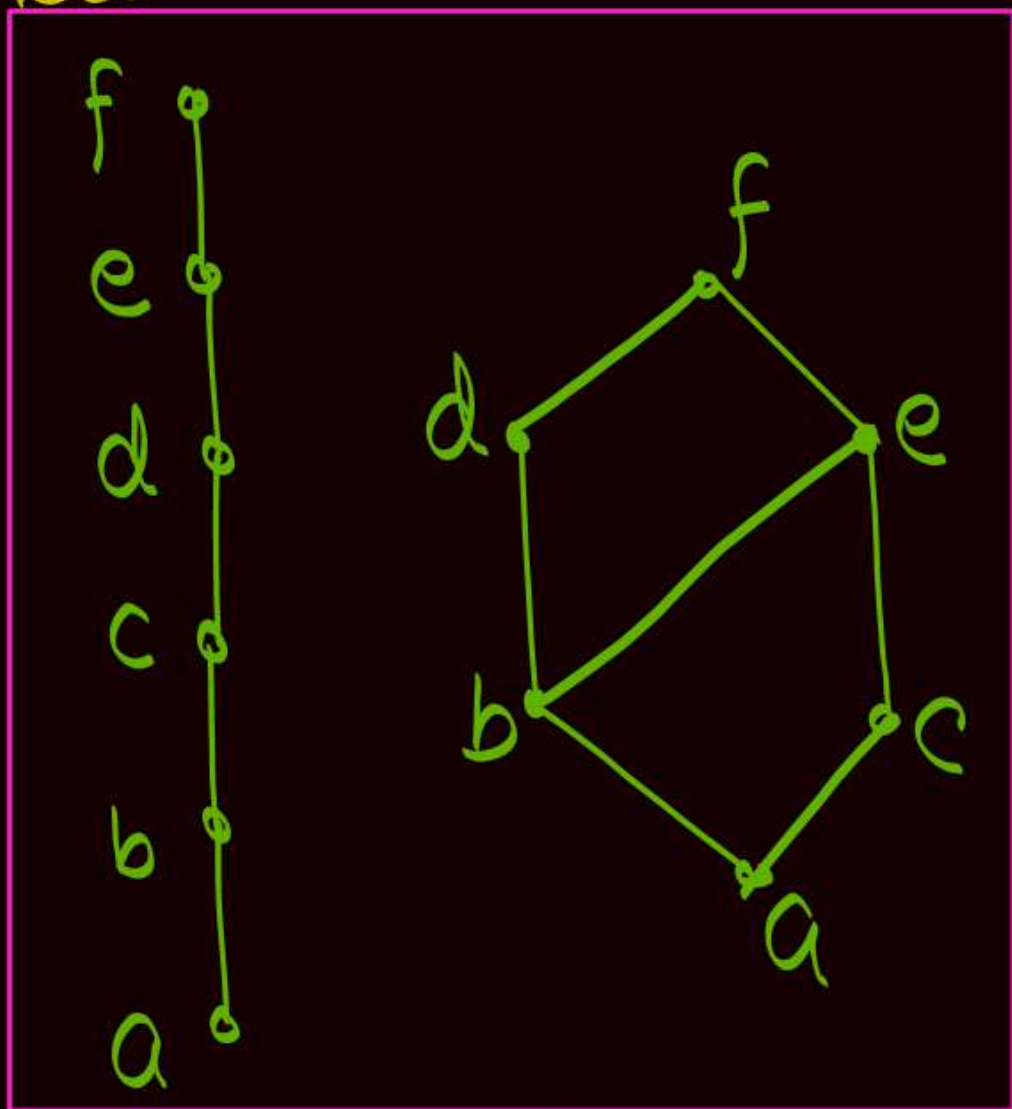
Note:- In a distributive lattice Every element has at most one complement.

Note:- In a lattice if any element has more than one complement, then that lattice can not be a distributive lattice.

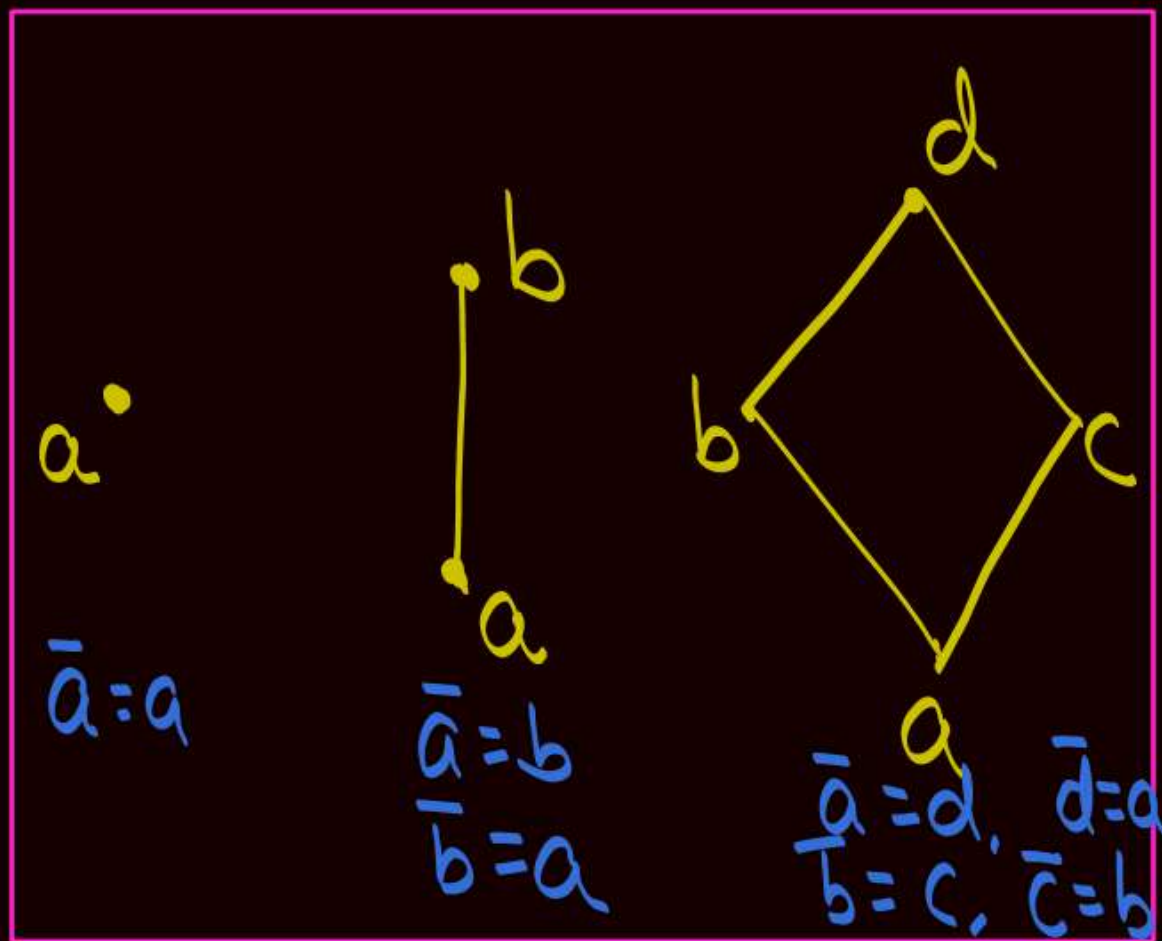
Note:- In a lattice if Every element has at most one complement, then that lattice may or may not be a distributive lattice.

Note:- In a lattice if Every element has exactly '1' complement, then that lattice is always a distributive lattice.

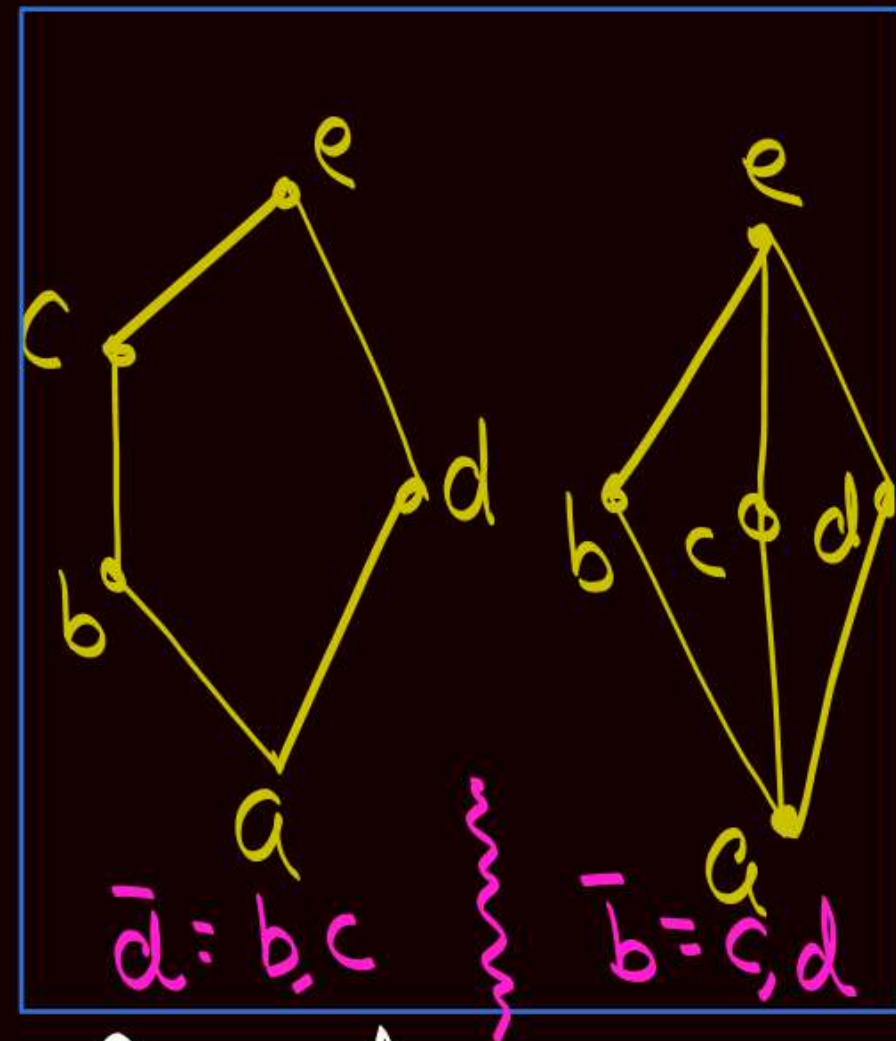
Note:-



Examples in which
Every element has
'0' or '1' Complement
or We need to check for
distributive property



Every element has exactly
one complement.
i. Above lattice are
distributive lattice



Some elements have
more than one complement
ii. Above lattice are
not distributive lattice

Note : $\rightarrow$ ① If we choose three elements from the same line of "Hasse diagram", then distributive property will always be satisfied by those elements.



② If Lattice is a TOSET, then distributive property will always be satisfied by the elements of that lattice

$\therefore$ A TOSET is always a distributive lattice

Note: -

These are power lattice w.r.t. TOSET
i.e. Distributive lattice



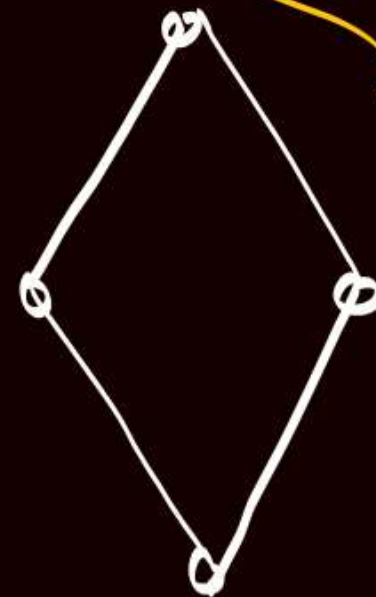
Hasse diagram
w.r.t. lattice
of size = 1

Hasse diagram
w.r.t. lattice
of size = 2

Hasse diagram
w.r.t. lattice
of size = 3



Hasse diagrams
w.r.t. lattice
of size = 4



Every element has
Exactly one
Complement
i.e. Distributive
lattice

Note: Any lattice with '4' or less elements is always distributive

Note:-

Hasse diagrams possible w.r.t lattice of size = 5

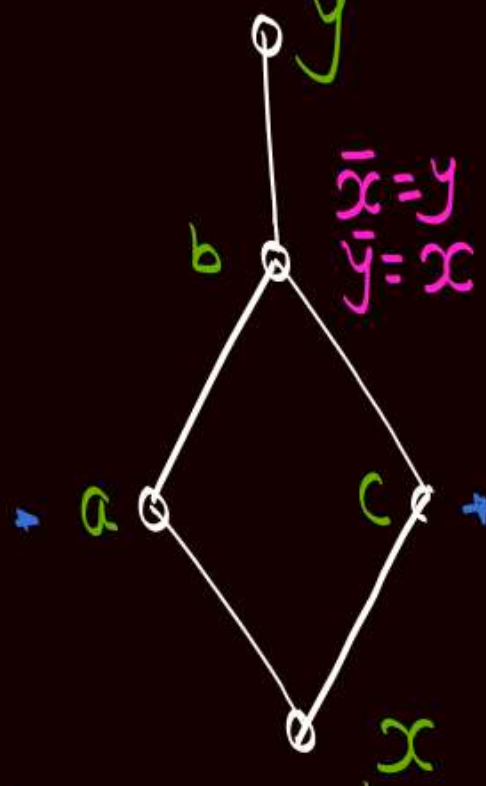
(i)



TOSET

• Distributive

(ii)

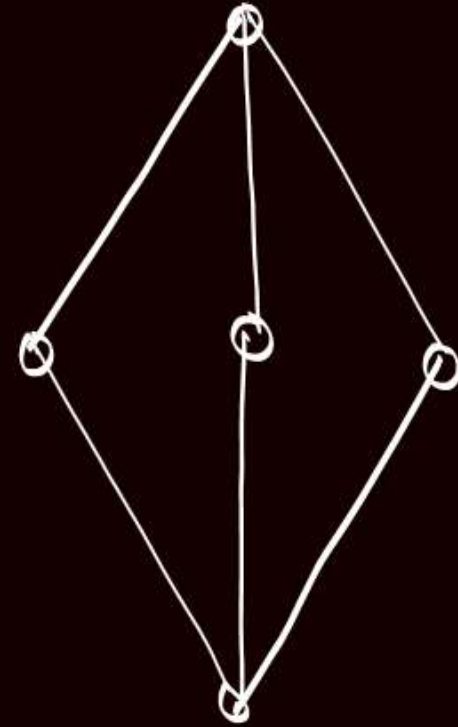
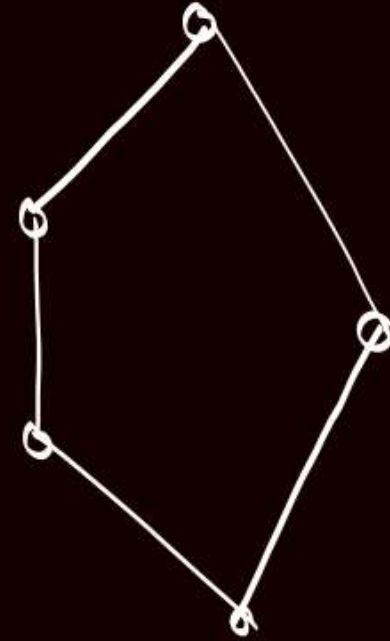


$\bar{x} = y$
 $\bar{y} = x$

(iii)



$\bar{x} = y$
 $\bar{y} = x$

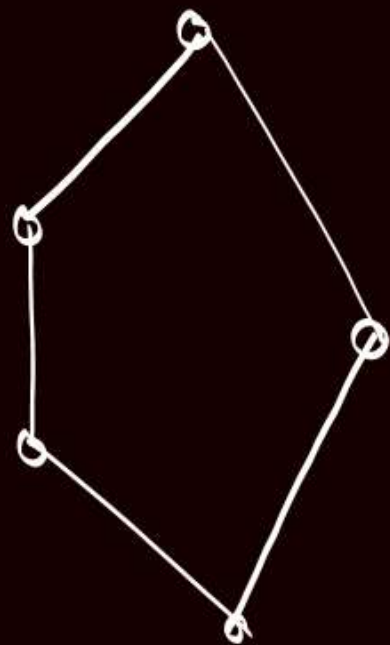


More than one Complement
for some elements
• Not distributive

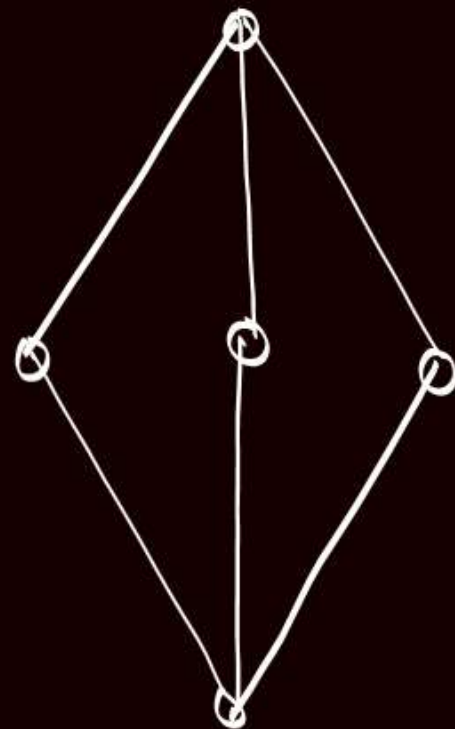
$\bar{a}$ = does not exist
 $\bar{b}$ = does not exist
 $\bar{c}$ = does not exist

$\bar{a}$ = does not exist
 $\bar{b}$ = does not exist
 $\bar{c}$ = does not exist

Every element has at most '1'
Complement, and
they are distributive



L_1^*



L_2^*

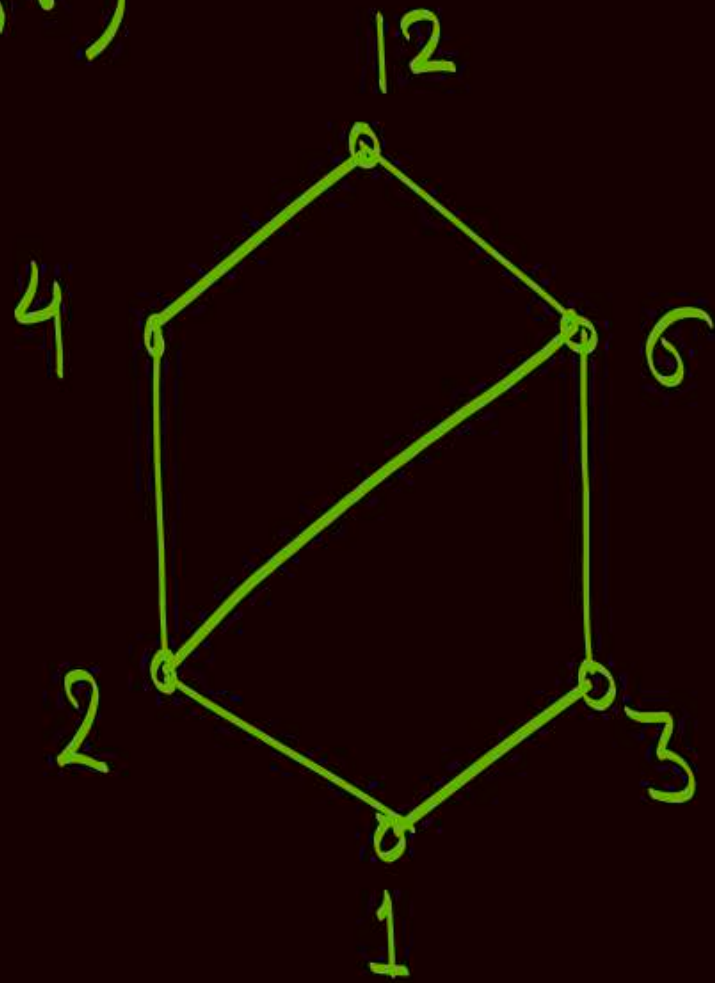
L_1^* and L_2^* are only two lattice of size = 5 that are not distributive

Note :- $\rightarrow$

A lattice L is not distributive
if and only if there exist a
sub-lattice of lattice L which
is isomorphic to L_1^* or L_2^*

Q:- Check whether the given lattice is distributive or not?

$(D_{12}, \div)$



① No. of elements > 4

$\therefore$ may or may not be distributive

② NOT a TOSET

$\therefore$ may or may not be distributive

③

$$\bar{1} = 12$$

$$\bar{12} = 1$$

$$\bar{3} = 4$$

$$\bar{4} = 3$$

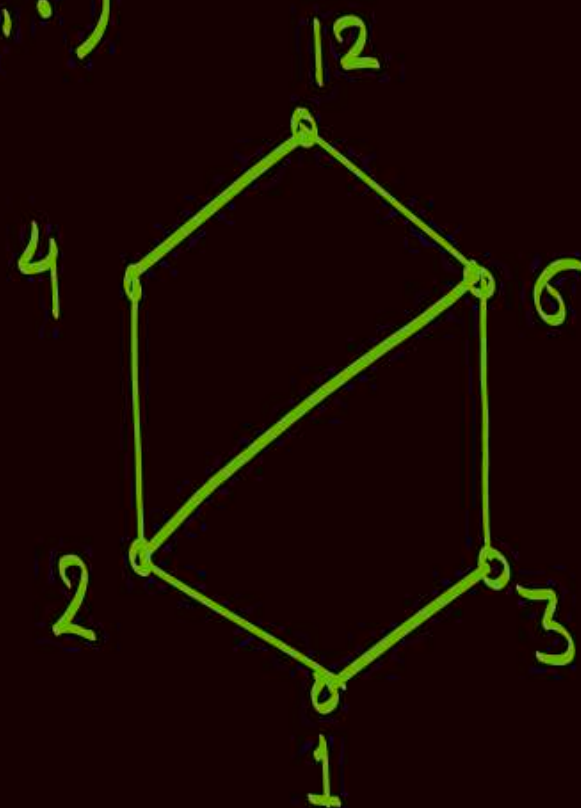
$$\bar{2} : \text{does not exist}$$

$$\bar{6} : \text{does not exist}$$

Every element has at most one complement
 $\therefore$ may or may not be distributive

Q:- Check whether the given lattice is distributive or not?

($D_{12}, \div$)



④ Do we have any sub-lattice of given lattice that is isomorphic to L_1^* or L_2^*

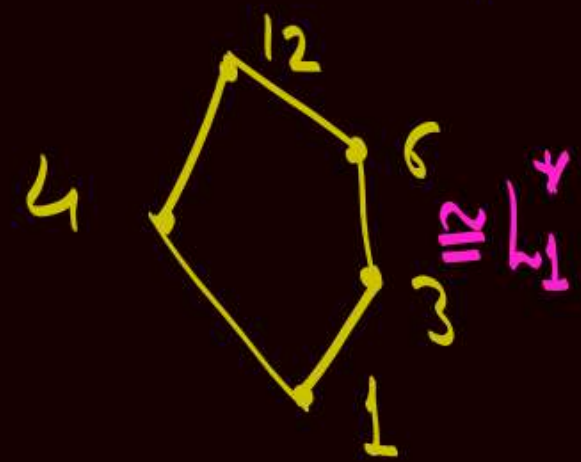
(i) we can not choose any subset of 5 elements s.t. Hasse diagram is isomorphic to L_2^*

(ii) $\{1, 3, 4, 6, 12\}$ and $\{1, 2, 3, 4, 12\}$ are two subsets of size = 5 s.t. Hasse diagram will be isomorphic to L_1^*

$\{1, 3, 4, 6, 12\}$

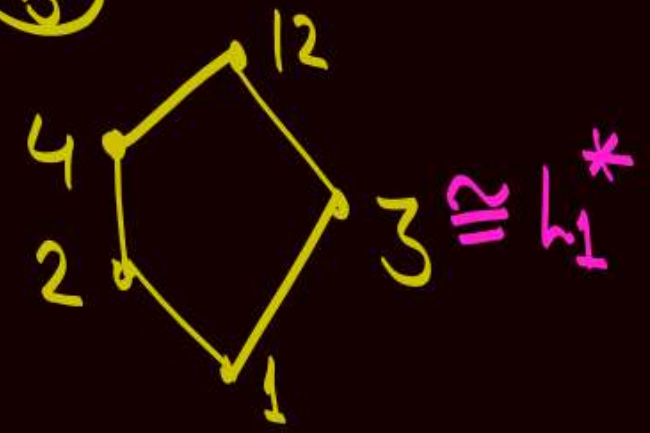
$\{1, 2, 3, 4, 12\}$

⑤



$\cong L_1^*$

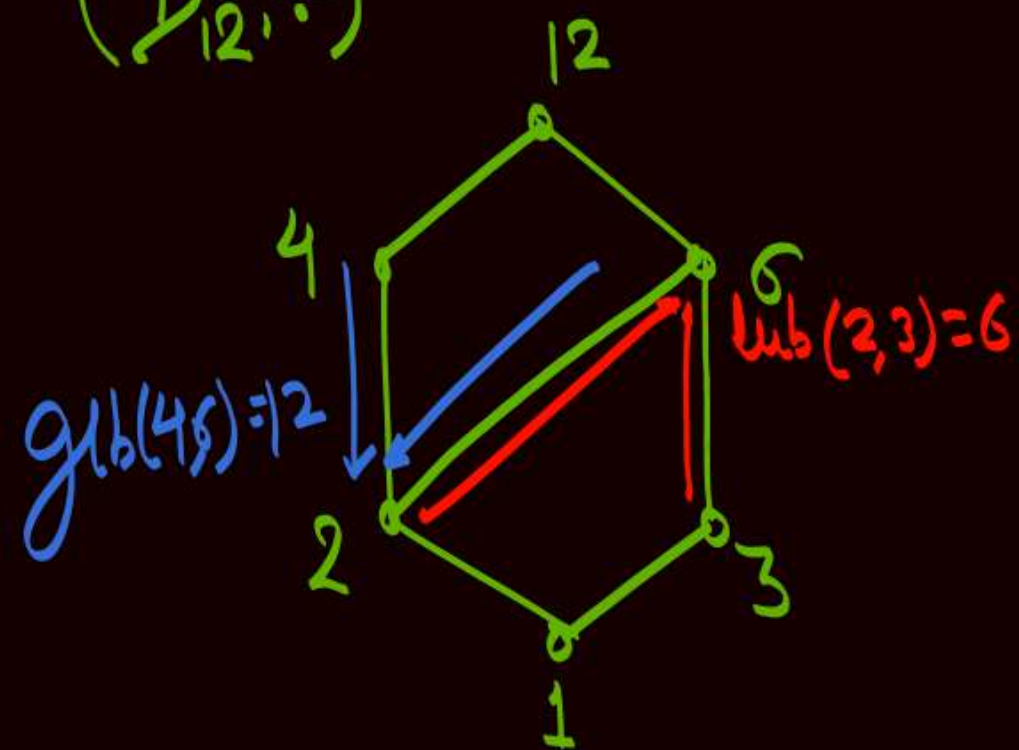
⑥



$\cong L_1^*$

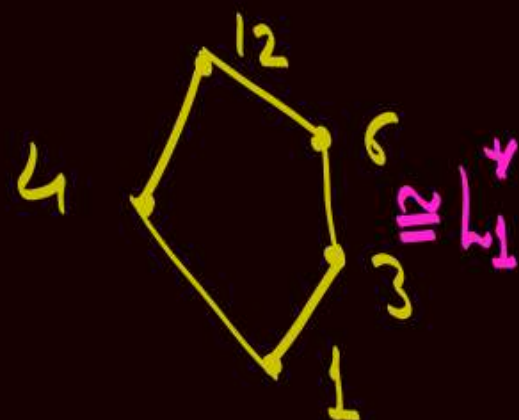
Q:- Check whether the given lattice is distributive or not?

(D_{12} :-)



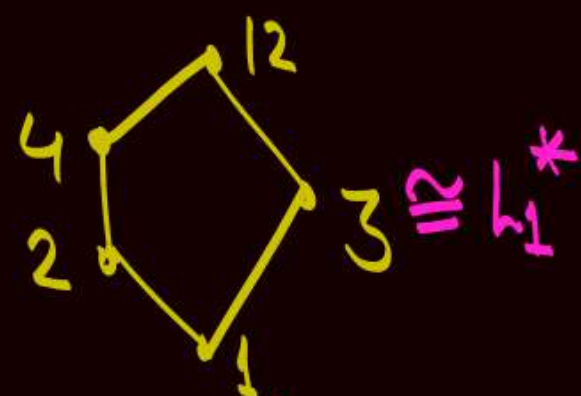
There is no other subset, s.t. Hasse diagram will be isomorphic to L_1^* or L_2^*

$\{1, 3, 4, 6, 12\}$



Hasse diagram wrt. subset $\{1, 3, 4, 6, 12\}$ is isomorphic to L_1^* , but it is not a sub-lattice of given lattice because $glb(4,6)$ differ

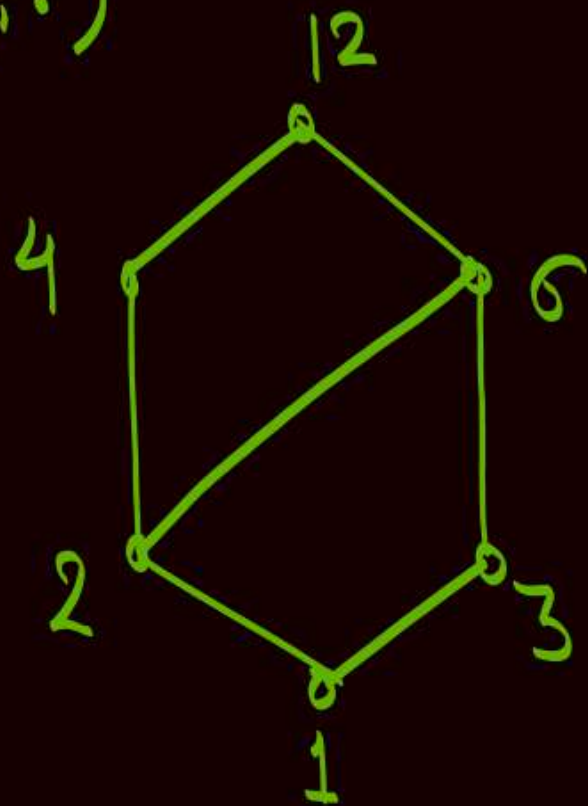
$\{1, 2, 3, 4, 12\}$



Hasse diagram wrt. subset $\{1, 2, 3, 4, 12\}$ is isomorphic to L_1^* , but it is not a sub-lattice of given lattice because $lub(2,3)$ differ

Q:- Check whether the given lattice is distributive or not?

($D_{12}, \div$)



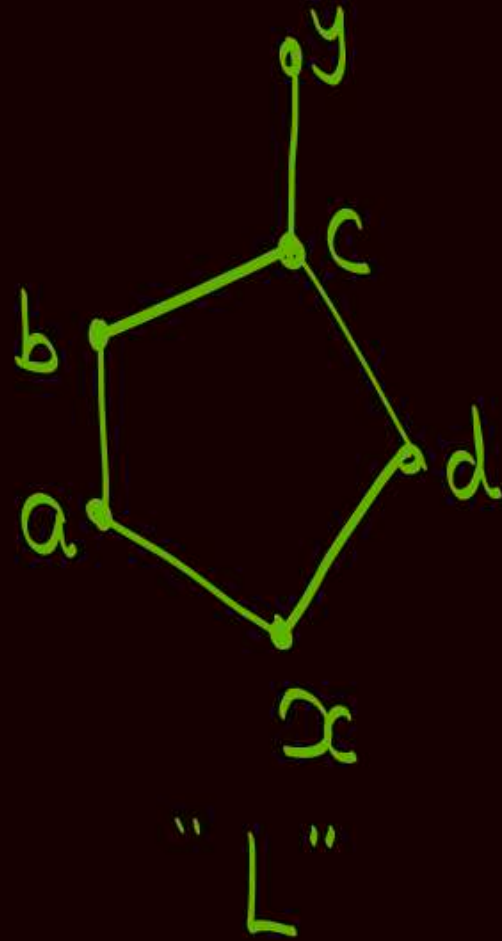
Q Do we have any sub-lattice of given lattice that is isomorphic to L_1^* or L_2^*

(i) There is no Sub-lattice of the given lattice that is isomorphic to L_1^*

And (ii) There is no Sub-lattice of the given lattice that is isomorphic to L_2^*

→ i.e, No sub-lattice, isomorphic to L_1^* or L_2^*
∴ given lattice is a distributive lattice

Q:- Check whether the following lattice is a distributive lattice or not?

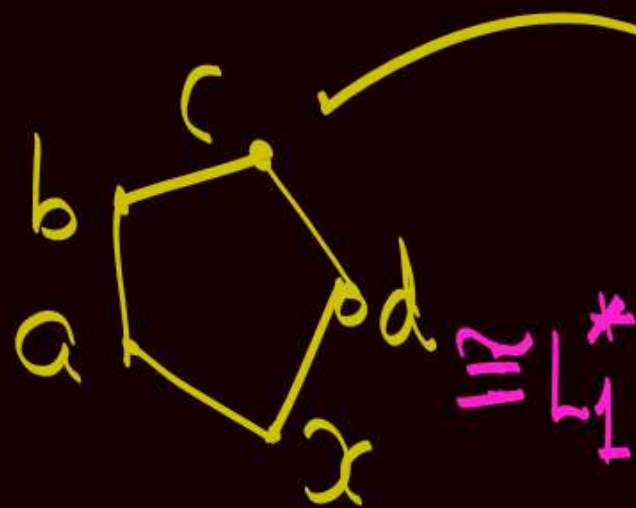


① No. of elements > 4
 $\therefore$ May or may not be distributive

② Not a TOSET
 $\therefore$ May or may not be distributive

③ $\bar{x} = y$
 $\bar{y} = x$
 $\bar{a}$ = Does not exist
 $\bar{b}$ = Does not exist
 $\bar{c}$ = Does not exist
 $\bar{d}$ = Does not exist

Every element has at most one complement
 $\therefore$ May or may not be distributive



④ $\{x, a, b, c, d\}$ is a sub-lattice which is isomorphic to L_1^*
 Hence Given lattice is not a distributive lattice



2 mins Summary



Topic

Distributive lattice

Slide

THANK - YOU